BONTHU VENKATA NAVEEN

AI & Automation Engineer | Generative AI Developer

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PROFILE SUMMARY

AI & Automation Engineer with **3+ years** of hands-on experience building production-grade **Generative AI** and **Agentic AI** systems. Specialized in building **Agentic AI systems**, **RAG architectures**, and **LLM-powered assistants** using **Azure OpenAI (GPT-4o)** and **Microsoft AutoGen**. Proven ability to integrate AI with enterprise platforms (**ServiceNow**, **Microsoft Graph API**), ship scalable **FastAPI** backends, and deliver measurable operational impact. Strong expertise in **Python automation**, FastAPI, and cloud-native AI solutions on Microsoft Azure.

KEY HIGHLIGHTS

- **Multi-Agent AI:** Designed and implemented a **3-agent Agentic AI system using Microsoft AutoGen**— Incident Analysis Agent, Meeting Agent, and Transcript Analysis Agent — to automate enterprise incident triage, stakeholder collaboration, and RCA generation.
- **LLM in Production:** Built a GPT-4o-powered AI using **Azure OpenAI** for real-time **RCA generation**, impact assessment, and resolution recommendations in a live enterprise environment.
- **RAG & Vector Search:** Implemented **RAG pipeline** using **Azure OpenAI embeddings** with **PostgreSQL (pgvector)** to retrieve historical incidents and runbooks, improving response accuracy and reducing hallucinations.
- **Automation at Scale:** Reduced manual patch validation effort by **50%**, saving **10+ engineer-hours/week** through self-healing infrastructure automation across Linux/Windows servers.
- **Enterprise Integrations:** Integrated AI workflows to **ServiceNow**, **Microsoft Graph API**, and **Microsoft Teams** for zero-touch stakeholder communication.

TECHNICAL SKILLS

- **Generative AI:** Azure OpenAI, GPT-4o, RAG, Prompt Engineering, Agentic AI, Multi-Agent Orchestration (Microsoft AutoGen), Vector Embeddings
- **Backend:** Python, FastAPI, REST APIs
- **Automation:** PowerShell, Shell Scripting, n8n (AI workflow automation & API integration)
- **Cloud:** Azure OpenAI Service, Azure AI Services, Azure Functions, AWS (Basic)
- **Enterprise:** Microsoft Graph API, ServiceNow API
- **Databases:** PostgreSQL (pgvector), MySQL,SQL
- **DevOps:** Docker, Git, GitHub
- **OS:** Linux, Windows

PROFESSIONAL EXPERIENCE

AI & Automation Developer **Capgemini India Pvt. Ltd., Bengaluru** **Dec 2022 – Present**

Project 1: AI-Powered Major Incident Management (MIM) Assistant

Client: Enterprise Operations | Tech Stack: Python | FastAPI | Azure OpenAI GPT-4o | Microsoft AutoGen | PostgreSQL (pgvector) | Microsoft Graph API | ServiceNow

Overview: Built an end-to-end **Agentic AI system** to automate major incident triage, root cause analysis, and stakeholder communication — replacing hours of manual coordination with an automated, LLM-driven pipeline.

- Architected **a multi-agent AI** system using **Microsoft AutoGen** with three specialized agents: **Incident Analysis Agent** (classifies incident severity and generates RCA), **Meeting Agent** (creates Microsoft Teams bridge meetings and notifies stakeholders using **Microsoft Graph API**), and **Transcript Analysis Agent** (extracts insights and action items from meeting transcripts). AutoGen coordinated agent interactions and managed the end-to-end workflow.
- **Integrated Azure OpenAI GPT-4o** for real-time incident analysis, automated impact assessment, resolution recommendations, and **Root Cause Analysis (RCA)** report generation.
- **Applied RAG pipeline** using **Azure OpenAI embeddings** with **PostgreSQL (pgvector)** to retrieve historical incident data and runbooks, improving response accuracy and reducing hallucinations.
- **Developed asynchronous FastAPI REST APIs** to expose AI workflows, integrate enterprise services, and handle concurrent incident processing.

- **Integrated Microsoft Graph API** and **ServiceNow APIs** to automate **Teams bridge creation**, stakeholder notifications, incident updates, and workflow automation.
- **Collaborated** across product, infra, and ops teams on solution design, UAT, deployment, and production support.
- **Impact:** Reduced **Mean Time to Resolve (MTTR) by ~40%** by automating incident triage, AI-driven knowledge retrieval, and RCA generation; approximately **30+ major incidents/month**, saving approximately **15+ engineer-hours/week** previously spent on manual triage, communication, and RCA documentation.

Project 2: Enterprise Infrastructure Automation Platform (ADC Bot Factory)

Client: Bayer | Tech Stack: Python | PowerShell | FastAPI | Linux | Windows | Git

Overview: Designed and delivered automation solutions for **server health monitoring**, **compliance validation**, patch verification, and auto-remediation across a large-scale Linux and Windows server estate.

- **Developed the Auto Heal Framework** — a self-healing engine that automatically detects and remediates infrastructure issues across **Linux and Windows**, reducing manual on-call intervention for common failure patterns.
- **Automated end-to-end patch validation** (pre- and post-patch checks), cutting manual effort by **50%** and saving **10+ hours/week**.
- **Developed reusable Python & PowerShell automation frameworks** for health monitoring, compliance validation, patch verification, and automated remediation across **hundreds of servers**.
- **Exposed automation via FastAPI REST APIs**, enabling integration with broader enterprise tooling.
- **Owned testing, debugging, and production support**; partnered with the client infrastructure team to improve automation reliability.

EDUCATION

B.Tech — Computer Science Engineering | Bonam Venkata Chala Mayya Engineering College

Aug 2018 – May 2022 | CGPA: 8.3 / 10

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals (AI-900)**
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